Thomas Szwarcer

tomszwarcer.github.io

Education

University of Oxford, Oriel College (MPhys Physics) 2021-2026

First Class degree classification.

Awards: Academic Scholarship prize for exam performance (Year 2 & Year 3); Collection Prize for performance in internal college exams (×6).
Final Degree Courses: Quantum mechanics, thermodynamics and statistical mechanics, electromagnetism, mathematical methods (Year 2); classical mechanics, fluids, condensed matter physics, nuclear and particle physics, symmetry and relativity, general relativity (Year 3); theoretical physics (including QFT), particle physics (Year 4).

Research Experience

CERN Summer Studentship 30/6/2025 - 19/9/2025

- Two new classes were added to the open source simulation toolkit Garfield++, and a simulation framework was written to include a model of the space charge effect in electron/ion drift simulations. This work enables more accurate simulations of gaseous and silicon detectors.
- **Contributions**: ability to drift ions in simulations with space charge; ability to use existing classes to drift charges arbitrarily without being confined to a grid; ability to pre-calculate the E-field on a 2D grid for efficiency, and interpolate to points between nodes; class added to approximate a charge distribution as a set of axisymmetric rings, with handling of divergences; line-by-line profiling of new and relevant existing code to significantly improve efficiency; ability to model resistive anodes, showing these are not responsible for superexponential gain observed in resistive Micromegas; successfully reproduced the expected gain-limiting effect of the space charge field in simulations; extension of framework to general detectors with less symmetry and multiple simultaneous avalanches.
- Progress regularly presented to the CERN GDD group. Seminar presentation given to the CERN EP-DT group. Project report written, and added as a DRD1 technical note. Work presented at the Oct. 2025 DRD1 collaboration meeting (Warsaw, Poland).
- Supervisors: Djunes Janssens, Heinrich Schindler.
- **Languages**: C++, Python, Bash. **Tools**: Garfield++, Git, HTCondor, lldb, Valgrind/Callgrind. **General**: Linux, techniques for code optimisation and efficiency, object-oriented programming, operational principles of gaseous detectors, presentation skills, report writing.

STFC Summer Studentship 1/7/2024 - 23/8/2024

- Investigation made into the use of Garfield++ and Magboltz for tracking the production of scintillation light in charge amplification simulations using Ar/CF4 mixtures. Application to the detector geometry and E-field configuration present in the MIGDAL experiment. A proof of concept scintillation tracking system was developed, and results compared to experiment data. Results indicated further work is required in understanding the mechanism of excitation transfer and scintillation in Ar/CF4 mixtures before it can be reproduced in simulations. Methods for reducing computation time of GEM drift simulations were investigated.
- Progress regularly presented to the collaboration. Seminar presentation given to researchers and other summer students. Progress regularly shared with the CERN GDD group. Work presented at a CERN DRD1 WG4 Working Meeting. Project report assembled.
- Supervisor: Pawel Majewski
- **Languages**: C++, Python, Bash. **Tools**: Git, HTCondor, Gmsh, Elmer FEM, Garfield++, Magboltz. **General**: Linux, OOP, operational principles of GEMs/TPCs, presentation skills, report writing.

Projects

MPhys Project

10/2025 - 4/2026

- Investigation into the pion Particle ID (PID) performance of the $K^0_S \to \pi\pi$ calibration sample for the LHCb experiment.
- Contributions to the data from obvious background processes were identified and removed. The sPlot technique was applied to unfold signal from the remaining background, making use of fits to the K^0_S invariant mass produced with RooFit.
- PID efficiencies and misidentification rates were computed. Two commonly used measures of PID confidence (DLL and ProbNN) were compared, revealing unexpectedly poor performance of ProbNN in the misidentification rates of π^+ as μ^+ or K^+. This could lead to a more informed choice of measure being used in future analyses.
- Results were compared to those of the existing pion PID calibration sample, $D^{*+} \to D^0\pi^+$. It was found that the $K^0_S \to \pi\pi$ calibration sample is suitable for use in place of (or to augment) the existing sample in the low-momentum region, where the latter suffers from poor purity and poorly-modelled background. The results obtained could lead to more reliable PID calibration for low-momentum pions.
- Project report assembled.
- Supervisor: Sneha Malde
- **Languages**: Python, Bash. **Tools**: ROOT, RooFit, HTCondor. **General**: Linux, techniques in analysis of particle physics data and large datasets, operational principles of the LHCb experiment, report writing.

Personal

Teaching	129 pages of LaTeX-formatted notes on quantum mechanics and linear algebra were created and distributed to students in years below. Developed a series of workshops on extracurricular physics to be delivered to A Level students at my local school, beginning November 2026. Experienced in giving private tuition (GCSE & A Level maths)
Languages	English (native speaker), Spanish ($\sim$ C1).
Music	Guitar, bass, drums. Production, songwriting, and performance.
Sport	Running, squash, gym.
General	Extensive experience in the service industry across various bar, hotel, and events roles.